



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目：信号与系统

开课单位：电子系

考试时长：2 小时

命题教师：陈晔、袁伟杰、刘凡

题号	1	2	3	4	5	6	7	8	9	10
分值	20	10	15	18	13	24				

本试卷共 (6) 大题，满分 (100) 分 (考试结束后请将试卷、答题本、草稿纸一起交给监考老师)

P1: Choose ALL correct answers ($20' = 2' \times 10$)

1) For an aperiodic DT signal $x[n]$, its DTFT $X(e^{j\omega})$ has the following properties _____.

- (a) continuous (b) discrete (c) periodic (d) aperiodic

2) A stable DT LTI system with impulse response $h[n]$ has its frequency response. This statement is _____.

- (a) true (b) false (c) depends on the properties of the input applied to the system

3) For a DT LTI system $y[n] = x[n] - x[n-1]$, its frequency response is equivalent to a _____.

- (a) low-pass filter (b) band-pass filter (c) high-pass filter (d) depends on the properties of $x[n]$

$$Y(e^{j\omega}) = X(e^{j\omega}) - e^{-j\omega} X(e^{j\omega}) \quad h[n] = \delta[n] - \delta[n-1]$$

$$H(e^{j\omega}) = \frac{1}{1 - e^{-j\omega}} \quad \omega = 0 \quad H = 1$$

$$\omega = 1$$

4) Choose all properties of an ideal CT low-pass filter _____.

- (a) It only has passband and stopband, but no transition between passband and stopband.
(b) It is a non-causal system.
(c) The gain of its passband is 1.
(d) The impulse response of an ideal CT low-pass filter is a rectangular pulse.

$$h(t) = \frac{\sin \omega_c t}{\pi t} \quad t < 0 \quad h(t) \neq 0$$

5) A CT signal $\cos(2\pi \times 60t)$ is sampled with a rate of $\omega_s = 160\pi$, and followed with an ideal low-pass filter with cutoff frequency 80π . The output signal is with the following frequency _____.

- (a) 120π (b) 40π (c) 160π (d) 200π

$$\omega_0 = 120\pi$$

6) For a DT signal $x[n]$ with sampling rate 100 Hz, downsampling to 50 Hz can be implemented with the following methods _____.

- (a) decimation with $N=2$
(b) convert $x[n]$ to a CT signal $x(t)$, sample the CT signal $x(t)$ with a rate of 50 Hz, and convert the sampled signal to DT sequence
(c) apply DT time expansion with $N=2$ to $x[n]$.
(d) pass signal $x[n]$ to an ideal DT low-pass filter with cutoff frequency $\pi/2$.

7) Which of the following can be used as the carrier signal in amplitude modulation? _____.

- ABCD ✓
 (a) complex exponential signal
 (b) ☒ sinusoidal signal
 (c) rectangular pulse train
 (d) sinc pulse train

8) For amplitude modulation with a sinusoidal carrier (frequency ω_c), choose all properties/requirements on its asynchronous demodulation _____.

- ABDE ✓
 (a) The message signal $x(t)$ should be always positive.
 (b) ☒ The message signal $x(t)$ are bandlimited $[-\omega_M, \omega_M]$, and $\omega_M < \omega_c$.
 (c) The envelope signal of the amplitude-modulated signal $y(t)$ approaches the message signal
 (d) ☒ Envelope detector could be designed with a diode and a low-pass filter.

9) When transmitting a PAM (pulse-amplitude modulation) signal, inter-symbol interference (ISI) happens at the receiver. Two methods can be used to against ISI, _____.

- AB ✓
 (a) ☒ use sinc pulse as the carrier signal in amplitude modulation
 (b) design an detector system at the receiver, and make sure that the overall frequency response of the LTI hold system, communication channel, and detector is an ideal low-pass filter
 (c) use asynchronous demodulation at receiver
 (d) use a larger pulse rate to generate the PAM signal

10) An LTI system is always a linear-phase system. This statement is _____.

- B ✓
 (a) true
 (b) false
 (c) depends on the properties of input signal applied to the system

P2 (10'=8'+2')

For a periodic train of impulses $y(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT - \frac{T}{2})$, where T is its fundamental period and

$\delta(t)$ is an impulse function, determine its Fourier transform $Y(j\omega)$, and sketch $Y(j\omega)$.

P3 (15'=3'+3'+4'+3'+2')

Consider a DT LTI system S with impulse response $h[n] = 0.2^n u[n]$, and with an periodic input (with period N)

$$x[n] = \cos\left(\frac{2\pi n}{N}\right).$$

- 1) Is this system a causal system or not? stable or not? With memory or memoryless?
- 2) Present the input signal $x[n]$ in Fourier series form.
- 3) What is the frequency response $H(e^{j\omega})$ of this system?
- 4) What is the system output $y[n]$?
- 5) If $N=4$, what is the phase difference between output $y[n]$ and input $x[n]$?

P4 (18'=6'+6'+6')

Determine the Fourier transform of the following signals

1) $x(t) = e^{-2|t-1|}$

2) $x(t) = \sin(\frac{\pi}{2}t) + \cos(t) + \delta(t)$

3) $x[n] = (\frac{1}{2})^{-n} u[-n-1]$

P5 (13'=6'+7')

Consider a causal and stable LTI system S whose input $x[n]$ and output $y[n]$ are related through the following difference equation:

$$y[n] - \frac{1}{6}y[n-1] - \frac{1}{6}y[n-2] = x[n]$$

1) Determine the frequency response $H(e^{j\omega})$ for the system S .

2) Determine the impulse response $h[n]$ for the system S .

P6 (24'=4'+4×5')

1) 以 $T=10^{-3}s$ 的采样周期进行采样后，请分别判断下列条件是否可作为完整还原原始信号的充分条件？其中 ω 代表圆频率。

a) 当 $|\omega| < 800\pi$ 时， $X(j\omega) = 0$

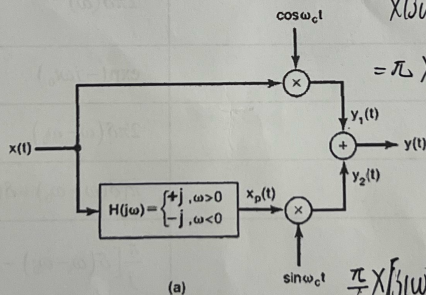
b) $x(t)$ 是一个实信号，并且当 $\omega > -1200\pi$ 时， $X(j\omega) = 0$

2) For the following system (a) with input $x(t)$ (with spectrum $X(j\omega)$ in (b)) and output $y(t)$,

i) sketch the 4 Fourier transform $Y_1(j\omega)$, $X_p(j\omega)$, $Y_2(j\omega)$ and $Y(j\omega)$ of signals $y_1(t)$,

$x_p(t)$, $y_2(t)$ and $y(t)$.

ii) describe the function of system (a) and its advantage.



$$X(j\omega) * \pi [\delta(\omega - \omega_c) + \delta(\omega + \omega_c)]$$

$$= \pi X(j(\omega - \omega_c)) + \pi X(j(\omega + \omega_c))$$

$$\frac{\pi}{j} X[j(\omega - \omega_c)] - \frac{\pi}{j} X[j(\omega + \omega_c)]$$

